

MOSTAFA ALI

Software Engineering • Cybersecurity & GRC • Data Analytics • 3D Web & Frontend • AI-Assisted Dev

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Computer Science Student | University of Guelph | Graduating April 2028

TECHNICAL SKILLS

- **Languages & Frameworks:** Python (Flask, Pandas, PyTest, ctypes, scripting), Java (OOP, Spring Boot fundamentals), C (Pthreads, Shared Libraries), SQL (PostgreSQL), JavaScript, TypeScript, HTML5, Bash/Shell, R (statistical analysis)
- **Frontend & 3D Engineering:** React 19, Next.js (App Router), Tailwind CSS, Three.js, React Three Fiber (R3F), Drei, GLSL Shaders (custom atmosphere rendering), Framer Motion (spring physics, scroll-linked animations), Bun, Vite, Vercel
- **Testing & Quality:** Black-box, Clear-box, UI, Regression & Integration Testing, JUnit, PyTest, Check (C), ctypes (Python-to-C integration), Valgrind, GDB, GitLab CI / GitHub Actions
- **DevOps & Infrastructure:** Docker, Kubernetes & Helm (actively learning), Jenkins (actively learning), Git, Linux/Unix (Advanced), Agile/Scrum
- **Data, Analytics & Security:** Power BI (dashboards & KPI reporting), Pandas, Matplotlib, Scikit-learn, Draw.io, NIST CSF & 800-53, ISO/IEC 27001, SOC 2, TLS/SSL, PKI, RBAC, PHIPA & GDPR, CompTIA Security+ (In Progress)
- **AI-Assisted Development:** GitHub Copilot, Cursor AI, Claude — used daily for code generation, test automation, algorithm optimization, documentation, and prototyping across all projects

TECHNICAL PROJECTS

Personal Portfolio Website | mostafaali.dev Mar 2026

React 19, Next.js, Three.js, React Three Fiber, GLSL Shaders, Framer Motion, Tailwind CSS, Bun, Vite, Vercel

- **3D Graphics & Custom Shader Engineering:** Built a high-performance 3D Earth at Night using Three.js and React Three Fiber with custom GLSL atmosphere shaders — optimized rendering pipeline using anisotropy and high-DPI scaling for a premium visual experience across all devices
- **Advanced Animation & Performance:** Implemented hyperspace scroll physics using Framer Motion spring animations and scroll-linked transitions — engineered smooth 60fps interactions balancing animation fidelity with bundle size and rendering performance
- **Full-Stack Deployment & Data Visualization:** Deployed on Vercel with Next.js App Router for server-side rendering and optimal SEO; integrated Recharts and Framer Motion data visualizations for interactive project dashboards

Architect & Developer | Risk & Security Control Management System (RSCMS) Feb 2026 – Present

Python, Flask, PostgreSQL, Java, Docker, PyTest, ctypes, Power BI, R, Cursor AI

- **Backend Architecture & GRC Compliance:** Designed a Flask REST API and normalized PostgreSQL schema mapping vulnerabilities to NIST 800-53 and ISO 27001 controls — producing SOC 2-ready audit trails and Power BI compliance dashboards tracking MTTR, MTTD, and control status for enterprise stakeholders
- **Algorithm-Optimized Data Pipeline:** Engineered a Python simulation generating 5,000+ synthetic incidents to stress-test data aggregation — analysed time and space complexity trade-offs to optimize query structures and pipeline performance at enterprise volume
- **ctypes Integration & Multi-layer Testing:** Wrote Python ctypes scripts validating the C shared library integration layer; applied black-box, clear-box, and UI testing strategies across the full stack to validate every layer independently

Lead Engineer | Sentry: AI-Driven Security Auditor (Team of 3) Jan 2026 – Feb 2026

C, Python, Java, JUnit, PyTest, ctypes, Docker, GitLab CI/CD, Linux

- **Systems Architecture & Concurrency:** Architected a multi-threaded C auditor using Pthreads — designed the concurrency model and telemetry pipeline algorithms for optimal throughput, operationalizing the NIST CSF Detect & Respond functions
- **CI/CD, Testing & Security:** Built JUnit/PyTest suites in a GitLab CI/CD pipeline maintaining 99% uptime; implemented RBAC authorization, TLS-aware data handling, and tamper-evident identity audit trails across the full stack

Match Predictor AI | Premier League Outcome Prediction Jan 2026

Python, Pandas, Scikit-learn, R, Random Forest Classifier

- **ML Pipeline & Algorithm Optimization:** Processed 1,400+ historical records with Pandas and R — engineered rolling average features and optimized a Random Forest classifier from 47% to 90% prediction accuracy through iterative hyperparameter tuning

Testing Suite — Graph & Game Engine | CIS*2750, University of Guelph Jan 2026

C, Python, Check Framework, ctypes, curses (UI), GitLab CI, Shared Libraries (.so)

- **Black-box, Clear-box & UI Testing:** Designed a Check test suite graded on defect detection across 9 hidden buggy library variants — applied all three testing methodologies including ctypes integration testing and curses UI validation; automated via GitLab CI

pipelines

CERTIFICATIONS

- PHIPA & GDPR Foundations | Microsoft Security Essentials (Professional Certificate) | (ISC)² Candidate | CompTIA Security+ (In Progress) | ISO 27001 & NIST 800-53 (Self-Study, Applied in RSCMS)

EDUCATION

Bachelor of Computing (Honours), Computer Science | University of Guelph [Graduating April 2028](#)

- Coursework: Analysis & Design of Computer Algorithms (CIS*3490), Software System Development (CIS*2750), Operating Systems (CIS*3110), OOP, Web Development (HTML5, TypeScript, JavaScript — in progress)